9 (a) Expand and simplify $(x - 2)(2x + 3)(x + 1)$

	$2x$	3
x	$2x^2$	$3x$
-2	$-4x$	-6

1 mark
Expand two brackets.

	$2x^2$	$-x$	-6
x	$2x^3$	$-x^2$	$-6x$
1	$2x^2$	$-x$	-6

1 mark
Multiply by the third bracket.

$2x^3 - x^2 - 6x + 2x^2 - x - 6$

$= 2x^3 + x^2 - 7x - 6$

1 mark
Collect like terms.

(3)

$$\frac{y^4 \times y^n}{y^2} = y^{-3}$$

(b) Find the value of n.

$4 + n - 2 = -3$

$n + 2 = -3$

$n = -5$

1 mark
Add powers when multiplying; subtract when dividing.

1 mark
Correct value of n.

(2)

(c) Solve $5x^2 - 4x - 3 = 0$
Give your solutions correct to 3 significant figures.

$a = 5, \quad b = -4, \quad c = -3$

$$x = \frac{-(-4) \pm \sqrt{((-4)^2 - 4 \times 5 \times (-3))}}{2 \times 5}$$

1 mark
Substitute into the quadratic formula.

$$x = \frac{4 \pm \sqrt{76}}{10}$$

1 mark
Simplify the expression.

$x = 1.27$ or $x = -0.472$ (3 s.f.)

1 mark
Both solutions correct.

(3)

(Total for Question 9 is 8 marks)

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